FDM Defect Detection – Team Work Plan (v2)

CSCI 6379 · Two-week sprint: Fri Oct 23 – Fri Nov 6, 2026 · Updated Oct 27 · Prepared by Hena Parween
Repository: github.com/Agya-Aboagye-Otchere/CSCI-6379-3D-Printing-Defect-Detection-and-Correction

1. Where we are

The project follows the proposal: classify six FDM defects (stringing, warping, layer shifting, under-extrusion, over-extrusion, bed adhesion) with transfer-learned MobileNet and ResNet (RQ1, RQ2), and map each predicted defect to corrective actions with a rule table that is checked against an independent reference (RQ3).

Item	Status
Real Roboflow data	Done. 6,132 images of the six classes from the 22,048-image export.
Near-duplicate frames removed	Done. Many images are frames of the same video. Similar frames are grouped into scenes and at most 30 images are kept per scene: 4,187 images remain.
70/15/15 split	Done, seed 42: 2,930 train / 631 val / 626 test. Whole scenes go to one split, so no scene appears in two splits. The images are in the repository (data/processed/).
Class weights	Done. Classes are imbalanced (layer shifting 108 vs. under-extrusion 1,421 images), so the weights matter.
EDA figures, 18 scenarios, independent reference	Done (figures/, docs/).
Models (RQ1, RQ2)	Done by Hena. MobileNetV2 and ResNet-50, two configurations × three seeds. Mean test macro-F1: MobileNetV2 64.3 ± 2.3%, ResNet-50 62.7 ± 0.9%; MobileNetV2 is 1.5× faster on the CPU with 11× fewer parameters. Harrish now reviews the results and writes them up.
Class audit	Waiting for Agya (due Oct 27). Sheet: data/rules/class_audit.csv.
Rule table (RQ3)	Waiting for Agya (due Oct 30). Template: data/rules/rule_table.json. The RQ3 code is ready.

Ignore the earlier Oct 15 handoff note and its 100% agreement result. The data at that time was synthetic (computer-drawn shapes, because the real zip had not been unpacked), and the rules being scored were written by the same person as the reference. Both problems are fixed; the old files are kept in archive/synthetic_data/.

2. Who does what

Person	Background	Owns	RQ	Report sections
Hena	Computer science	Data pipeline, model training, simulation, RQ3 agreement evaluation	RQ3 evaluation	Data; Experimental Setup; RQ3 results
Harrish	Computer science	Review of the model training and results; write-up of Methods and RQ1/RQ2	RQ1, RQ2	Methods; Results for RQ1 and RQ2
Agya	Manufacturing	Class audit, defect definitions, correction rule table, error interpretation	RQ3 rules	Background; Related Work; rule discussion

3. Tasks by person

Agya (manufacturing): defect knowledge and rules

- **Class audit (by Tue Oct 27).** Look at about 50 random images per class in data/processed/train/. Warping vs. bed adhesion and under- vs. over-extrusion are the likeliest mix-ups. Write problems to data/rules/class_audit.csv with the columns image_path, issue, suggested_class (a class name, or drop). Commit and push the file; Hena re-runs the cleaning with your findings.
- **Rule table (by Fri Oct 30).** Fill in data/rules/rule_table.json. For each defect give a primary action, up to two secondary actions, the direction (INCREASE, DECREASE, ENABLE or CHECK), a typical range and at least one citation (Jin et al. 2019, Brion et al. 2022, De La Rosa et al. 2024, or troubleshooting guides). Use the parameter names in the file, then commit and push it.
- **Add Harrish to the repository** (Settings → Collaborators), as was done for Hena.
- **Independence rule:** do not open src/independent_reference.py or docs/INDEPENDENT_REFERENCE_TABLE.md until the rule table is submitted. RQ3 only means something if the two were written separately.
- **Error interpretation (Nov 4–5).** Explain the models' confusion matrices in physical terms.
- **Write:** Background, Related Work, and the rule-table discussion.

Harrish (computer science): model review and RQ1/RQ2 write-up

The training was moved to Hena on Oct 26 to save time. Harrish does not need to train anything; everything below is in the GitHub repository (clone it, then git pull before each session).

- **Review the training (by Thu Oct 29):** read src/train.py and check that both models really use identical settings, that the test split is used only once, and that the class weights and augmentation match the plan. Send Hena anything that looks wrong.
- **Review the results (by Thu Oct 29):** reports/model_metrics_mobilenet.json and model_metrics_resnet.json, figures/confusion_matrix_*.png and figures/training_curves_*.png. Look for over-fitting and for classes with low recall.
- **Seed variation:** reports/seed_runs_summary.json has the mean and standard deviation over three seeds for both configurations; the final configuration was chosen on validation, not test.
- **Write Methods (Section 6) and RQ1/RQ2 Results (Section 7) by Wed Nov 4.** Start from the generated draft reports/MODEL_RESULTS.md: it has the settings table, the comparison table (accuracy, macro precision/recall/F1, ms per image) and per-class recall. Turn it into prose and answer RQ1 and RQ2 directly.
- **Discussion input (Nov 5):** the MobileNet vs. ResNet trade-off between accuracy and speed.

Hena (computer science): data and simulation

- **Data (done).** Real data loaded, cleaned, near-duplicate frames grouped into scenes, split by scene, class weights computed, EDA figures regenerated.
- **Models (done Oct 26).** MobileNetV2 and ResNet-50 trained with identical settings, two configurations × three seeds; metrics, figures and the test-set predictions are in reports/ and figures/.
- **Code on GitHub (done Oct 27).** Code, split images, reports and figures pushed to the team repository.
- **Apply the class audit (Oct 27–28)** and re-run the pipeline. If the split changes, retrain both models and push the new results.
- **Scenario evaluation (Oct 30–31):** score the rule table against the reference on the 18 scenarios: primary-action agreement, any-action overlap and Cohen's kappa.
- **End-to-end simulation (Nov 3–4):** replay each model's test predictions through the rules and score every recommendation against the reference for the image's true class. Reports agreement, agreement when the system acts, and the abstention rate below 65% confidence, for MobileNet vs. ResNet.
- **Write (Nov 2–6):** Data, Experimental Setup and RQ3 results, with figures.

4. Two-week schedule

Dates	Hena	Harrish	Agya
Oct 23–27	Share this plan and the repo; train both models (done)	Clone the repo	Class audit → class_audit.csv; add Harrish to the repo
Oct 28–30	Apply audit; retrain if the split changed	Review training code and results	Rule table complete
Nov 2–3	Scenario evaluation; RQ3 figures	Methods draft	Background and Related Work draft
Nov 4–5	End-to-end simulation for both models	RQ1/RQ2 results write-up; discussion input	Confusion-matrix interpretation
Nov 6	Merge the report; final check	Proofread the merged report	Rule discussion section

5. Handoff files

File	Written by	Used by	Due
data/manifests/data_split_manifest.csv	Hena	Harrish	Ready
data/rules/class_audit.csv	Agya	Hena	Oct 27
data/rules/rule_table.json	Agya	Hena	Oct 30
reports/predictions_*.csv, model_metrics_*.json, MODEL_RESULTS.md	Hena	Harrish (review, write-up)	Ready
reports/AGREEMENT_RATE_EVALUATION.json	Hena	Everyone (report)	Nov 4

All files are exchanged through the GitHub repository: `git pull` before starting, then `git add`, `git commit` and `git push` after a change. Edit only your own files to avoid conflicts.